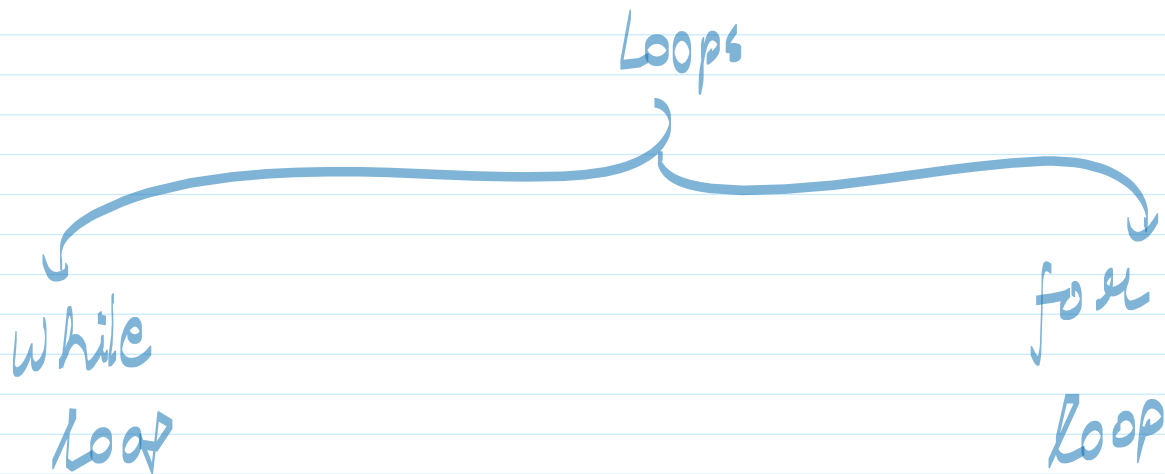


Loops :-

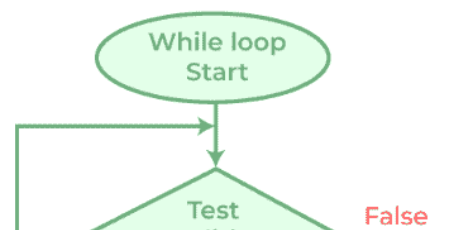


loop



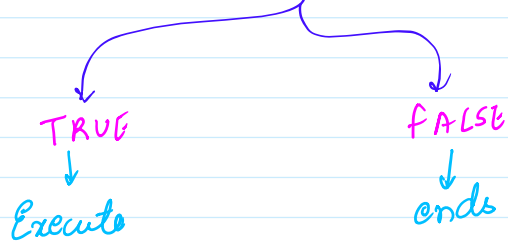
While loop :-

- 1) Loop start (Initialization)
- 2) Condition checks



1) Loop starts (Initialization)

2) Condition checks

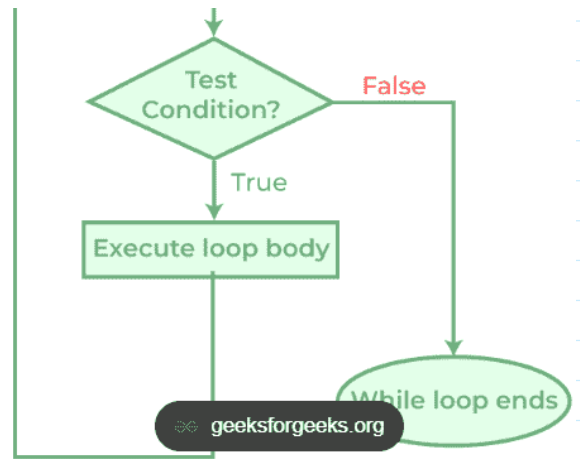


```
i=1 # Loop Initialization  
while i<=1000: → Condition  
    print("Kiaan")  
    i+=1  
    } loop body
```

$i = 7 \neq 1001$

$7 < 1000 \rightarrow \text{True}$

$1001 \leq 1000 \rightarrow \text{false}$



1. Kiaan
2. Kiaan
3. Kiaan
4. Kiaan
5.
6. ...

.. 1000. Kiaan

Iteration:-

➤ One cycle of successful loop is called iteration.

While loop with Else:-

```
condition=False  
while condition==True:  
    print("Sophie")  
else:  
    print("Azaan")
```

Range function :-

`range(start : stop: step)`

$\text{range}(5) \rightarrow 0, 1, 2, 3, 4$

range(0, 5): 0 1 2 3 4

Hande(1,11):

Observation: Last no. does not count

For loop :-

```
for i in range(1,1001):  
    print("Sophie")
```

```
i=1 # Loop Initialization
```

```
while i <= 1000:
```

```
print("Sophie")
```

 $i += 1$

automatic

```
for i in range(1,11,2):  
    print("Sophie")
```

0

✓ ✓ ✓ ✓ ✓

1 2 3 4 5 6 7 8 9 10

i

Sum of N natural number :

Ex- $1 + 2 + 3 + 4 + 5 + \dots + N = \frac{N(N+1)}{2}$

$N = 5$

$Sum = 0$ ← Initially.

while

Dry Run:-

$i = 1$

while $i \leq 5$:

print ("Kiaan")

$i += 1$

Step 1:

$i = 1$

$i \leq 5$

$1 \leq 5$ ✓

1. Kiaan

~~$i = 2$~~

Step 2:

$i = 2$

$i \leq 5$

$2 \leq 5$ ✓

2. Kiam

i = 3